CS432: Databases

Introduction to Databases

Instructor
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Lecture no.
2

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What is a Database

A database is an organized collection of data, generally stored and accessed electronically from a computer system.

Disclaimer..

- This course is NOT
  - a tutorial on using a specific databases
  - a tutorial on SQL
  - a course on database implementation

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  - a tutorial on using a specific databases
  - a tutorial on SQL
  - a course on database implementation
- But it is about learning
  - the foundations of database design
  - some SQL and relational algebra
  - optimization techniques in database design
  - managing large databases

Course Contents (Racap)

- Introduction to RDBMS.
- Structured Query Language (SQL).
- Relational Algebra, Entity-Relationship Model, Relational Database Design
- Storage and File Structure
- Application Development
- Indexing and Hashing
- Query Processing, Query Optimization - Transactions (Serializability and Recoverability)
- Concurrency Control
- Recovery Systems
- Introduction to no-SQL databases

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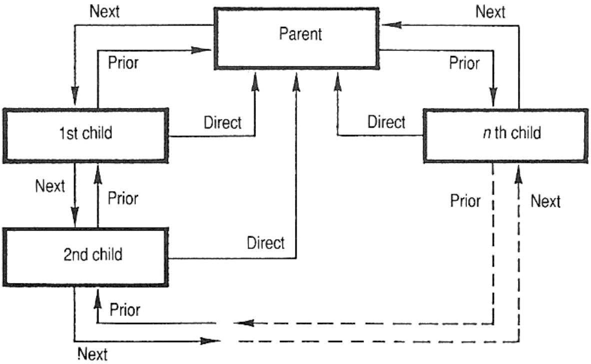
Sometimes we even don't see them. Can you come up with some examples?

The paradigm shift..

The term "data-base" was coined around 1962.

- Navigational DBMS (1960')



A closed chain of records in a navigational database model (e.g. CODASYL), with ***next pointers***, ***prior pointers*** and ***direct pointers*** provided by keys in the various records.

Source: Wikipedia

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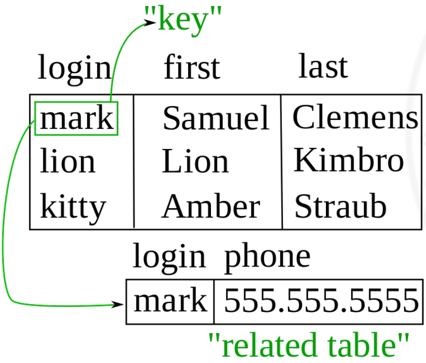
- Relational DBMS (1970’)



Edgar Codd, “**A Relational Model of Data for Large Shared Data Banks**”

Source: Wikipedia

Relational Databases: An example

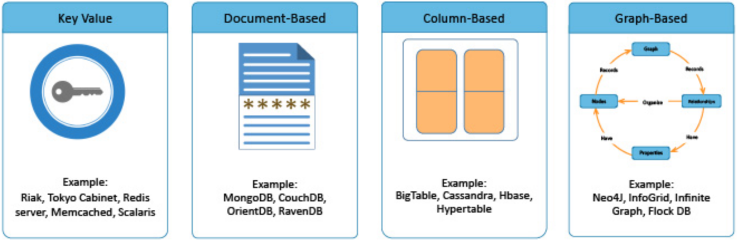
ID	*name*	*dept_name*	*salary*
22222	Einstein	Physics	95000
12121	Wu	Finance	90000
32343	El Said	History	60000
45565	Katz	Comp. Sci.	75000
98345	Kim	Elec. Eng.	80000
76766	Crick	Biology	72000
10101	Srinivasan	Comp. Sci.	65000
58583	Califieri	History	62000
83821	Brandt	Comp. Sci.	92000
15151	Mozart	Music	40000
33456	Gold	Physics	87000
76543	Singh	Finance	80000

Source: Silberschatz, Korth, Sudarshan — Database System Concepts

The paradigm shift..

The term "data-base" was coined around 1962.

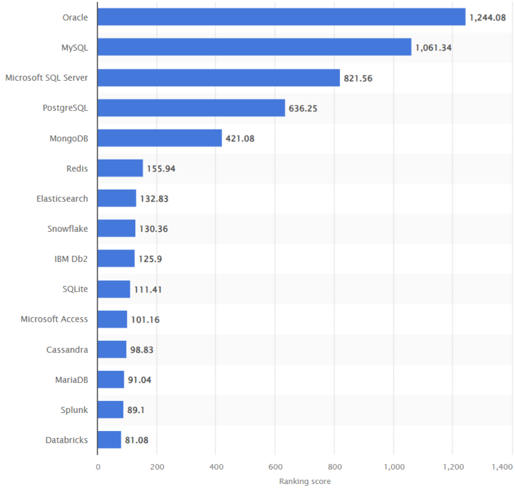
- NoSQL and NewSQL (2000')



NoSQL Database Types

Feature	Document	Key-Value	Columnar	Graph
Data Model	JSON-like documents	Key-Value pairs	Columns instead of rows	Nodes & Relationships
Best Use	Semi-structured data	Fast lookups & caching	Analytics & big data	Relationship-heavy data
Query Perf.	Moderate	Fast	High (Analytics)	Optimized (Links)
Schema	Flexible	Dynamic	Semi-structured	Schema-less
Scalability	Horizontal	High horizontal	Highly scalable	Link-based scaling
Examples	MongoDB, CouchDB	Redis, DynamoDB	Cassandra, HBase	Neo4j, Neptune

DBMS Rankings (June 2024)



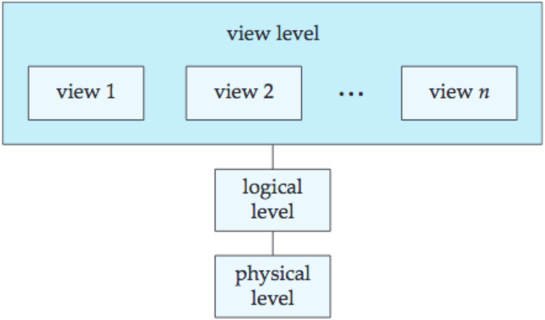
Source: statista.com

DB-Engines (source): https://db-engines.com/en/ranking

Data abstraction

Not all database-system users are not computer trained, we need to hide the complexity

- Physical level
- Logical Level
- View level



Source: Silberschatz, Korth, Sudarshan — Database System Concepts

Physical level

- How the data are actually stored in the system.
- Describes complex low-level data structures.

Logical level

- Describes what data are stored in the database.
- What relationships exist among those data.
- Describes the entire database in terms of a small number of relatively simple structures (e.g. Tables).
- **Physical data independence:** User of the logical level does not need to be aware of the complexity at physical layer.

View level

Not everything (the entire database) should be visible to everyone...

- Users need to access only a part of the database.
- Simplifies the interaction of the users with the system.
- Many views for the same database.

An example

type instructor = **record**
ID:**char**(5);
name:**varchar**(20);
deptname:**varchar**(20);
salary:**numeric**(8,2);
end;

Some more record types:

- department: dept name, building, and budget
- course: course id, title, dept name, and credits
- student: ID, name, dept name, and tot cred

How does these can be described at different levels of abstraction?

Instances and Schemas

An analogy to a program written in a programming language.

Schema

Variable declarations corresponds to Schema

Instances

Value of a variable corresponds to Instances

Instances and Schemas

Schema example

type instructor = **record**
ID:**char**(5);
Name:**varchar**(20);
DeptName:**varchar**(20);
Salary:**numeric**(8,2);
end;

Instance example

ID	Name	DeptName	Salary
C0383	Yogesh	CSE	80000.00

Schema Types

- **Logical Schema:** the overall logical structure of the database.
- **Physical schema:** the overall physical structure of the database.

Programmers/Database administrators construct applications by using the logical schema

Data Models

A collection of tools to describe:

- Data
- Data relationships
- Data semantics
- Consistency constraints

Categories of data models

- **Relational Model:** Tables (relations), Columns (fields or attributes)
- **Entity-Relationship Model:** Real objects (entities) and relationships among these objects.
- **Object-Based Data Model:** Extension of ER with encapsulation, methods (functions), and object identity.
- **Semi-structured Data Model:** XML

Some older models include: Network model and Hierarchical model

Acknowledgments/Contributions

- Some of the images utilized in these slides are subject to copyright - Abraham Silberschatz, Henry Korth, and S. Sudarshan. Database System Concepts. 6th Edition, McGraw-Hill Education
- Contributor 2024-25, 2025-26 - Yogesh K. Meena, IIT Gandhinagar